Supporting Information

UniTopo: A Topology-Guided Unified Multi-Scale Representation Learning Framework for Molecular Property Prediction

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1 Detailed Molecular Graph Specifications and Feature Sets

1.1 Node and Edge Feature Breakdown

Table S1: Statistics of node and edge features in three molecular graphs.

Graph	Node features	Edge features	Node feature types	Edge feature types
Atom (A)	79	10	Atom properties	Bond properties
Junction tree (J)	83	6	Edges, rings and intersection atoms with the number of each atom	Types of connected nodes
Functional group (F)	115	20	Predefined types of edges, rings and functional groups	Types of connected nodes and number of intersections

1.2 Substructural Fragmentation Rules

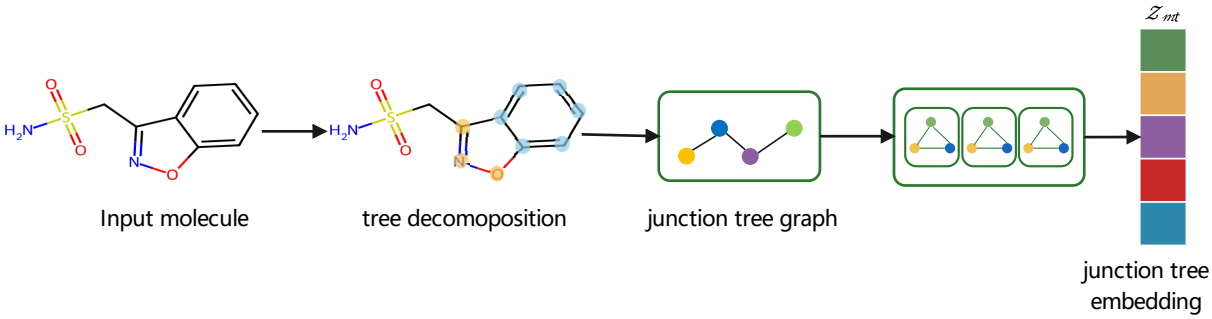


Figure S1: The molecular tree decomposition module. The input molecule is decomposed into a junction tree [1] whose nodes and edges are encoded into the junction-tree embedding $\mathbf{z}_{mt}$ by a GIN-based branch.

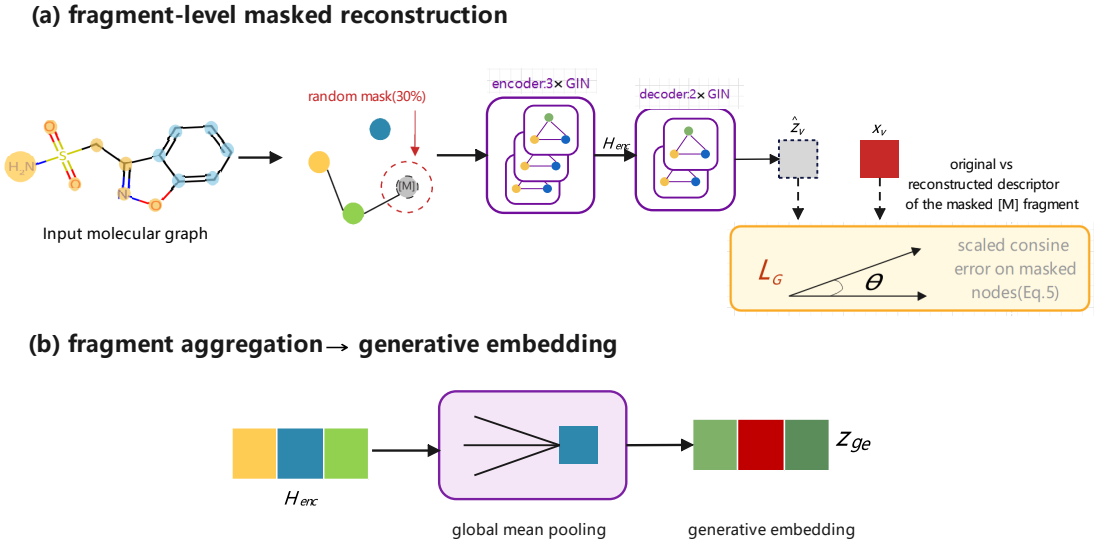


Figure S2: The fragment-based molecular generation module. (a) Fragment-level masked reconstruction with a graph masked autoencoder [2], in the spirit of hierarchical masked autoencoders such as Hi-GMAE [3], supervised by the scaled cosine error $\mathcal{L}_G$ of Equation (5). (b) Global mean pooling of the encoder features yields the fragment embedding $\mathbf{z}_{fm}$.

2 Detailed Architectural and Mathematical Formulations

2.1 The Three View Encoders

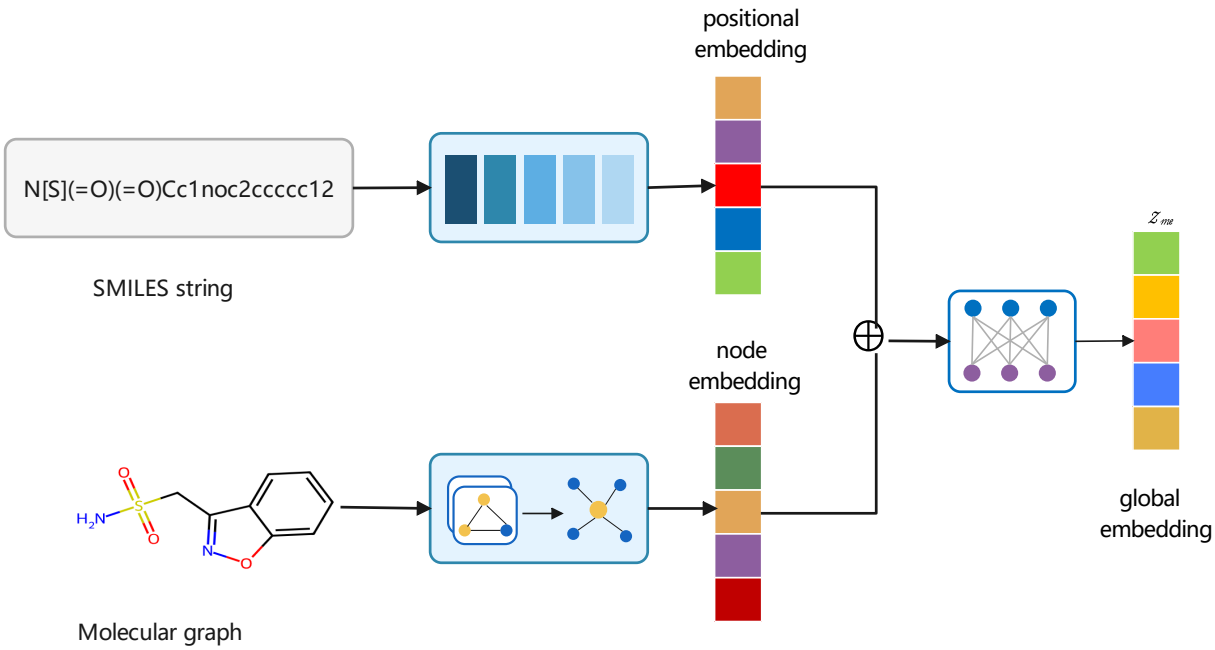


Figure S3: The molecular graph embedding module. The SMILES string [4] and the virtual-node-augmented molecular graph, encoded by GIN layers [5], are combined in parallel and fused before the Transformer encoder; sum-pooling yields the graph-level embedding $\mathbf{z}_{me}$.

2.2 Fragment-Based Masked Autoencoding (GMAE) Reconstruction

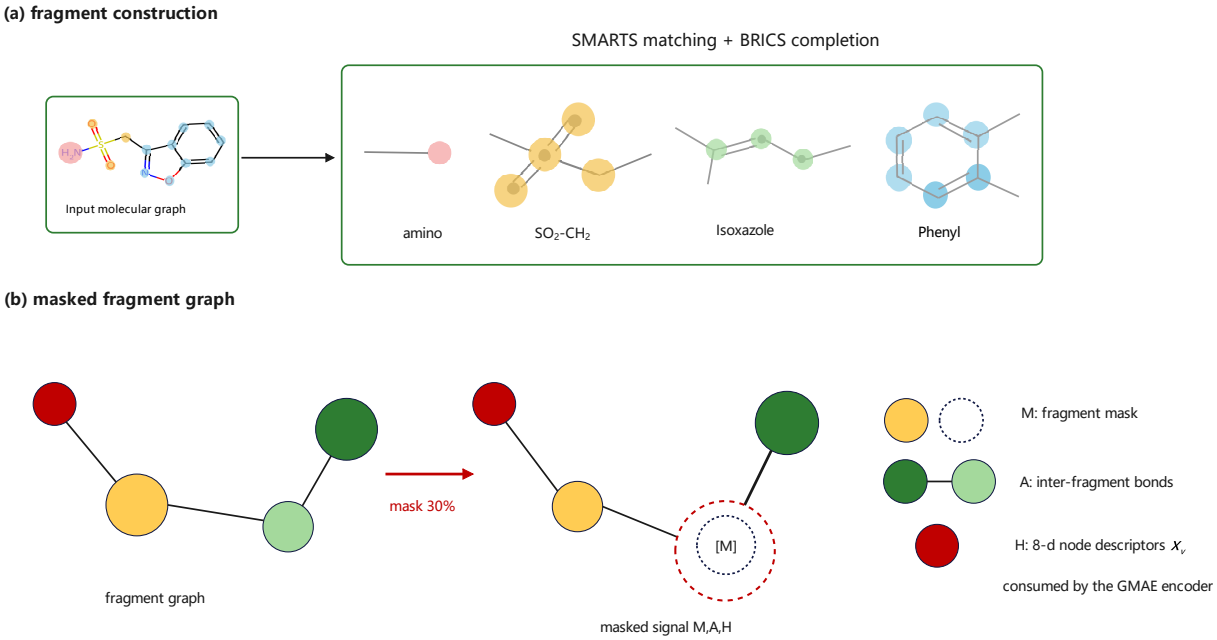


Figure S4: Construction of the masked fragment graph. Functional-group matching with BRICS completion [6] turns the molecule into a single fragment graph whose nodes and edges are subsequently masked; the asterisk marks a bond cleaved by BRICS and left open.

2.3 Element-Feature Fusion (EFF) and Contrastive Alignment

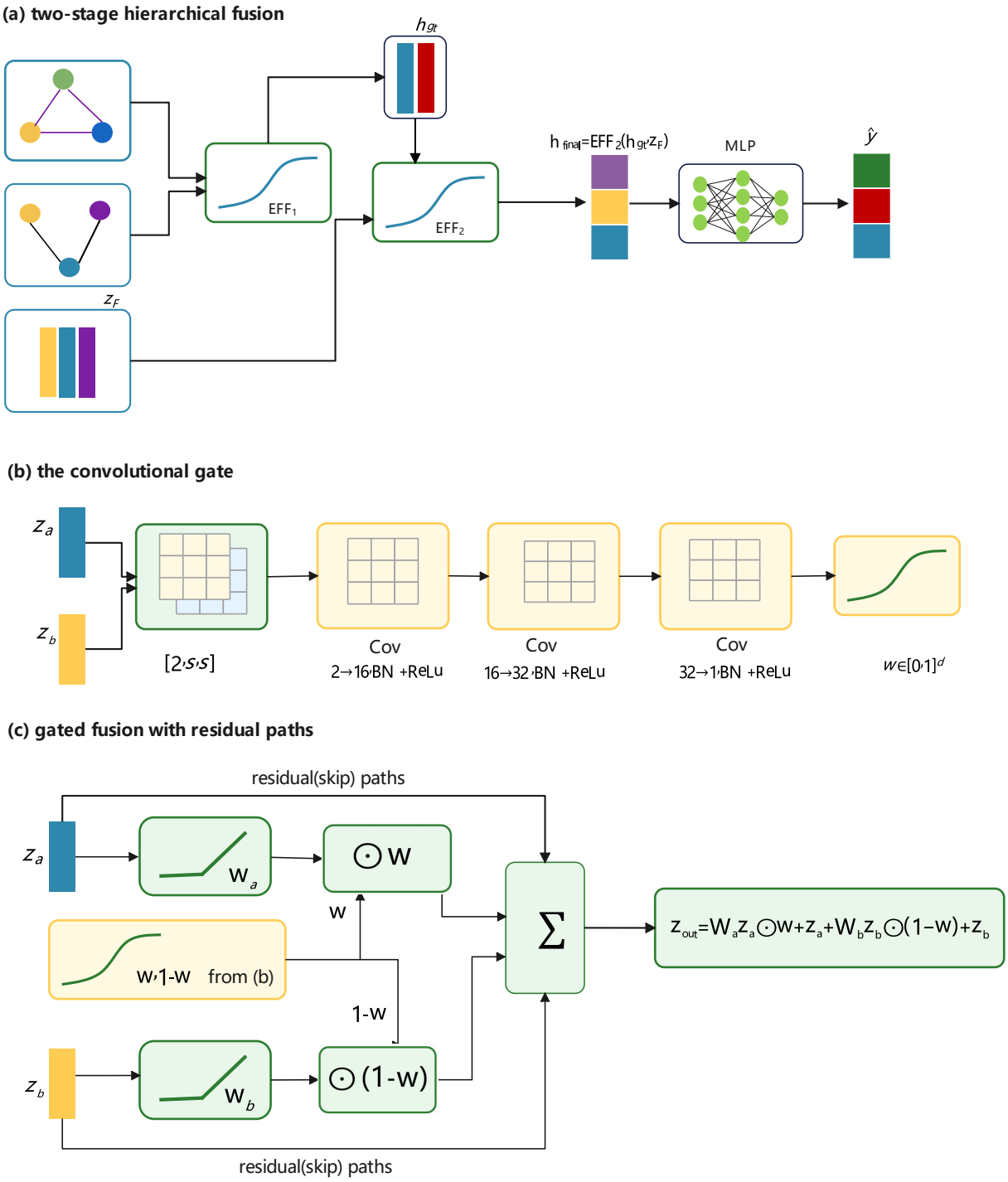


Figure S5: Element-Feature Fusion. (a) The two stages of Equation (6): EFF_1 combines the atomic and junction-tree views into $\mathbf{h}_{gt}$, and EFF_2 combines that result with the functional-group view. (b) The gate of one EFF block (Equation 7). (c) The gated fusion of Equation (8), with residual paths carrying the unweighted views into the sum.

3 Benchmark Dataset Statistics and Experimental Configurations

3.1 Training and Hyperparameter Setup

Table S2: Setting parameters for the UniTopo model.

Parameter	Description	Value or range
Encoder layer	GIN number of encoder layers	4
Heads of attention	Attention span of Transformer	6
Batch size	Batch size of the input data	64, 128, 256
Epochs	Epochs	100
Pre-trained lr	Initial learning rate, pre-training stage	1e-3 – 5e-3
Downstream lr	Initial learning rate, downstream stage	1e-3 – 5e-4
Pre-trained embedding	Embedded dimension, pre-training stage	300
Downstream embedding	Embedded dimension, downstream stage	256
Dropout	Drop rate for both stages	0.3

3.2 Downstream Benchmark Details

Table S3: Performance of the UniTopo model on four molecular graph configurations, classification tasks (ROC-AUC, mean ± s.d. over six seeds). All four configurations are fine-tuned from the same pre-trained checkpoint under the protocol of the capacity-controlled comparison (six seeds, 40 fine-tuning epochs), and the full configuration (A+J+F) corresponds to the UniTopo model. A = Atomic Graph, J = Junction Tree Graph, F = Functional Group Graph.

Model	BACE	BBBP	ClinTox	HIV	SIDER	Tox21
A	88.81 ± 1.41	86.42 ± 0.99	87.85 ± 3.11	80.21 ± 1.78	65.19 ± 2.62	71.49 ± 1.24
A+J	88.60 ± 1.75	85.53 ± 2.41	86.77 ± 1.95	78.35 ± 2.25	64.71 ± 1.04	72.87 ± 1.08
A+F	89.59 ± 1.42	86.40 ± 1.93	90.48 ± 1.67	79.92 ± 2.95	62.24 ± 3.54	69.43 ± 1.14
A+J+F	**95.8** ± 0.60	**98.7 ± 0.20**	**97.5 ± 0.80**	**87.9 ± 0.40**	**69.3 ± 0.30**	**82.5 ± 1.10**

Table S4: Performance of the UniTopo model on four molecular graph configurations, regression tasks (RMSE/MAE, mean ± s.d. over six seeds). The QM7 column reports the MAE in kcal/mol (the units of the dataset, matching the baselines of Table 3); the FreeSolv, ESOL and Lipo columns are in the standardised label units of the reduced-budget protocol, and QM8 in the units of its dataset. All four configurations are fine-tuned from the same pre-trained checkpoint under the protocol of the capacity-controlled comparison (six seeds, 40 fine-tuning epochs), and the full configuration (A+J+F) corresponds to the UniTopo model. A = Atomic Graph, J = Junction Tree Graph, F = Functional Group Graph.

Model	FreeSolv	ESOL	Lipo	QM7	QM8
A	0.7752 ± 0.3151	**0.3331** ± 0.0285	0.6271 ± 0.0172	57.897 ± 3.387	0.0302 ± 0.0008
A+J	**0.6198** ± 0.1183	0.3415 ± 0.0118	0.6237 ± 0.0207	56.127 ± 1.741	0.0294 ± 0.0004
A+F	0.6850 ± 0.2785	0.3525 ± 0.0241	0.6308 ± 0.0147	59.419 ± 2.609	0.0291 ± 0.0006
A+J+F	0.9032 ± 0.4874	**0.3349 ± 0.0235**	**0.6189 ± 0.0162**	**46.400 ± 0.900**	**0.0105 ± 0.0002**

4 Supplementary Analyses of the Pre-Training Objectives

4.1 Sensitivity to the Loss Balancing Factor

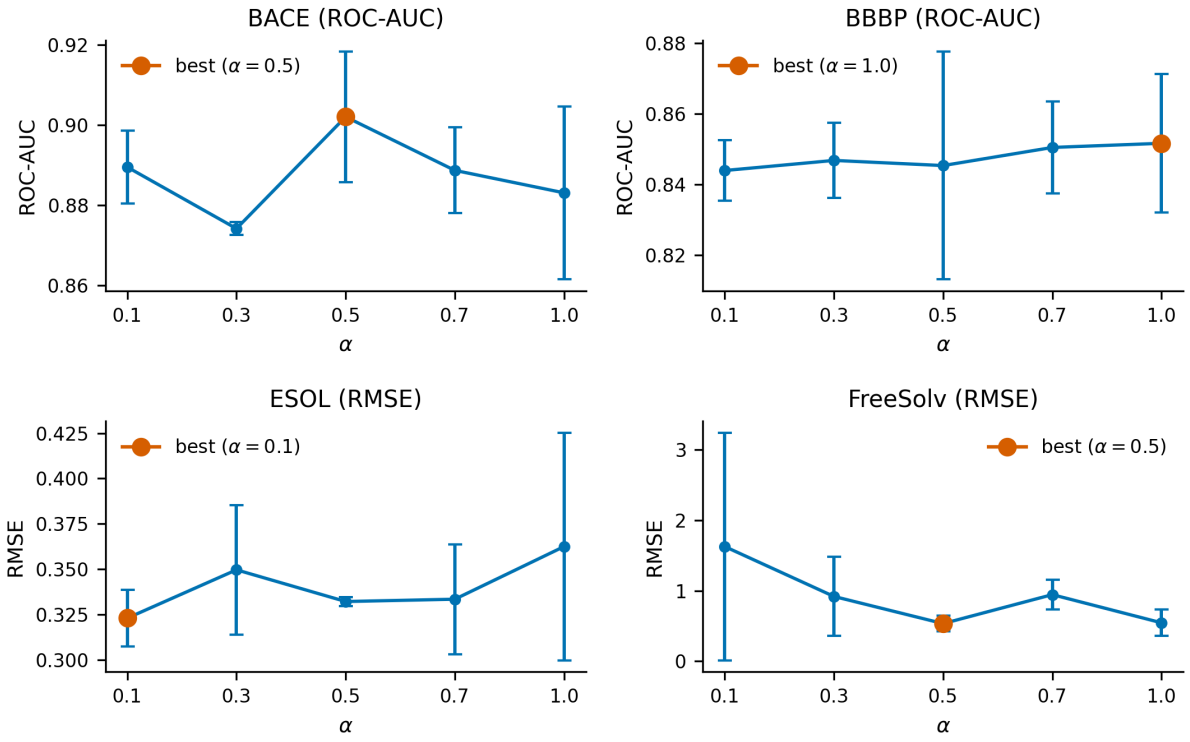


Figure S6: Sensitivity of downstream performance to the loss balancing factor α: ROC-AUC on BACE and BBBP (top) and RMSE on ESOL and FreeSolv (bottom); the orange marker in each panel marks that endpoint's best setting.

4.2 Alignment Diagnostics: Consistency without Collapse

Table S5: Alignment diagnostics for the pre-trained encoders, 600 molecules per dataset.

Dataset	Within-molecule similarity (A|J / A|F / J|F)	Margin (A|J / A|F / J|F)	View-identity probe	Permutation null	Effective rank (A / J / F)
BBBP	+0.114 / −0.025 / −0.006	+0.090 / +0.008 / +0.008	1.000	0.327	58.2 / 66.9 / 81.6
ESOL	+0.083 / −0.007 / −0.004	+0.025 / +0.005 / −0.007	1.000	0.339	51.3 / 46.1 / 60.9

Table S6: Dependence between the three view embeddings, 600 molecules per dataset.

Dataset	View pair	CKA	Predictive R^2 (forward)	Predictive R^2 (reverse)
BBBP	atomic / junction tree	0.412	0.674	0.524
BBBP	atomic / functional group	0.664	0.694	0.610
BBBP	junction tree / functional group	0.337	0.362	0.257
ESOL	atomic / junction tree	0.464	0.861	0.654
ESOL	atomic / functional group	0.671	0.782	0.764
ESOL	junction tree / functional group	0.458	0.623	0.607

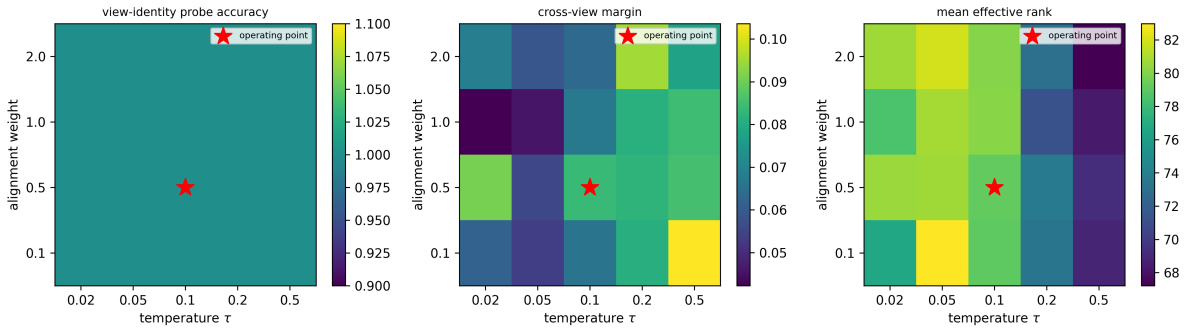


Figure S7: Alignment sensitivity sweep over $\tau \in 0.02, 0.05, 0.1, 0.2, 0.5 \times$ weight $\in 0.1, 0.5, 1.0, 2.0$, at a reduced pre-training budget for the trend across the grid. Left: view-identity probe accuracy; centre: cross-view similarity margin; right: mean effective rank. The red star marks the operating point ($\tau = 0.1$, weight $= 0.5$) used throughout the paper.

4.3 Gradient Interaction between the Objectives

Table S7: Gradient interaction between the two pre-training objectives, measured over a complete uninterrupted twenty-epoch run. A positive cosine means the two gradients point in the same direction.

Quantity	Value
Cosine over all parameters, per-epoch mean	+0.003 (range −0.010 to +0.013, positive in 16/20 epochs)
Cosine over shared parameters, per-epoch mean	+0.006 (range −0.018 to +0.029, positive in 16/20 epochs)
Shared parameters	729,699 / 7,259,080 (10.1%)
Gradient mass in the shared set	33.4%

5 Supplementary Characterisation Results

5.1 Junction-Tree Decomposition by Ring-System Complexity

Table S8: Junction-tree decomposition by ring-system class, over four benchmarks spanning physiological (BBBP), biophysical (BACE), physical-chemical (ESOL) and quantum-chemical (QM8) endpoints. Clusters per molecule is the mean number of junction-tree nodes; atom coverage is the fraction of heavy atoms assigned to a cluster.

Dataset	Ring-system class	Molecules	Clusters per molecule	Atom coverage
BBBP	acyclic	99	1.05	1.000
BBBP	mono-ring	251	1.92	1.000
BBBP	fused	96	1.78	1.000
BBBP	multi-ring	492	3.34	1.000
BBBP	polycyclic-fused	1085	2.84	1.000
BBBP	macrocyclic	16	4.06	1.000
BACE	acyclic	5	1.00	1.000
BACE	mono-ring	11	1.91	1.000
BACE	fused	11	1.64	1.000
BACE	multi-ring	926	4.27	1.000
BACE	polycyclic-fused	556	3.77	1.000
BACE	macrocyclic	4	2.50	1.000
ESOL	acyclic	317	1.00	1.000
ESOL	mono-ring	386	1.57	1.000
ESOL	fused	87	1.32	1.000
ESOL	multi-ring	167	2.63	1.000
ESOL	polycyclic-fused	171	1.99	1.000
QM8	acyclic	3643	1.00	1.000
QM8	mono-ring	10202	1.39	1.000
QM8	fused	3569	1.21	1.000
QM8	multi-ring	1217	2.12	1.000
QM8	polycyclic-fused	3155	1.43	1.000

Table S9: Per-molecule test error by ring-system class. For QM8 the metric is mean absolute error; for BBBP it is the mean absolute difference between the predicted probability and the label, averaged over the six runs and then over molecules. Strata with fewer than 20 molecules are greyed out as unreliable.

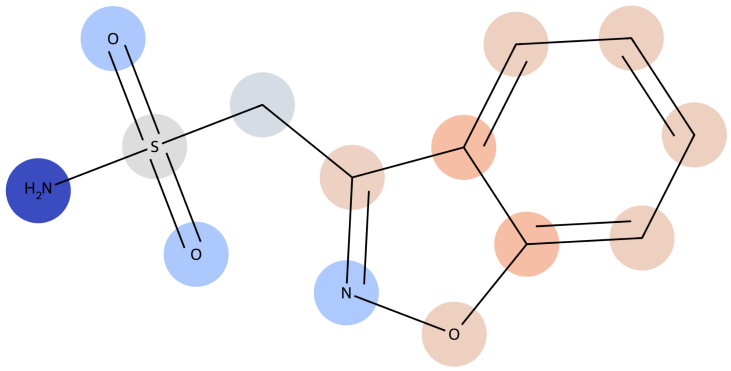
Dataset	Ring-system class	Molecules	Clusters per molecule	Atom coverage	Test error
QM8	acyclic	136	1.00	1.000	**0.0091**
QM8	mono-ring	1593	1.59	1.000	0.0143
QM8	fused	343	1.41	1.000	0.0110
QM8	multi-ring	28	2.11	1.000	**0.0193**
QM8	polycyclic-fused	80	1.29	1.000	0.0093
BBBP	acyclic	8	1.00	1.000	0.0474 (n < 20)
BBBP	mono-ring	9	1.89	1.000	0.0663 (n < 20)
BBBP	fused	19	1.63	1.000	0.4217 (n < 20)
BBBP	multi-ring	51	3.35	1.000	0.2687
BBBP	polycyclic-fused	114	2.88	1.000	0.1293
BBBP	macrocyclic	4	4.00	1.000	0.0103 (n < 20)

5.2 Characterisation of the Scaffold Splits

Table S10: Structural separation between training and test molecules. Maximum ECFP4 Tanimoto similarity between each test molecule and the training set, plus the fraction of test molecules whose Bemis-Murcko scaffold occurs in training, per benchmark. Averages over the eleven benchmarks: median max-Tanimoto 0.474 ± 0.106, near-analogue fraction 62.4%.

Dataset	Scaffolds shared with train (%)	Median max-Tanimoto	10th percentile	Test molecules above 0.4 (%)
BBBP	3.9	0.455	0.266	61.5
BACE	0.7	0.746	0.633	99.3
ClinTox	8.8	0.453	0.221	56.8
SIDER	8.3	0.473	0.250	61.8
Tox21	0.0	0.455	0.335	67.8
HIV	3.7	0.388	0.270	45.4
FreeSolv	0.0	0.370	0.270	37.5
ESOL	36.0	0.500	0.291	78.1
Lipo	0.2	0.564	0.309	72.9
QM7	0.7	0.391	0.320	45.0
QM8	6.2	0.421	0.355	60.0

(A) Zonisamide (BBBP #349, permeable)
ClogP = 0.616 TPSA = 86.2 Å² sulfonamide -1.689 isoxazole +0.558 benzene ring +1.223



(B) Isoniazid (BBBP #279, impermeable)
ClogP = -0.315 TPSA = 68.0 Å² hydrazide N-N -1.729 pyridine +0.444

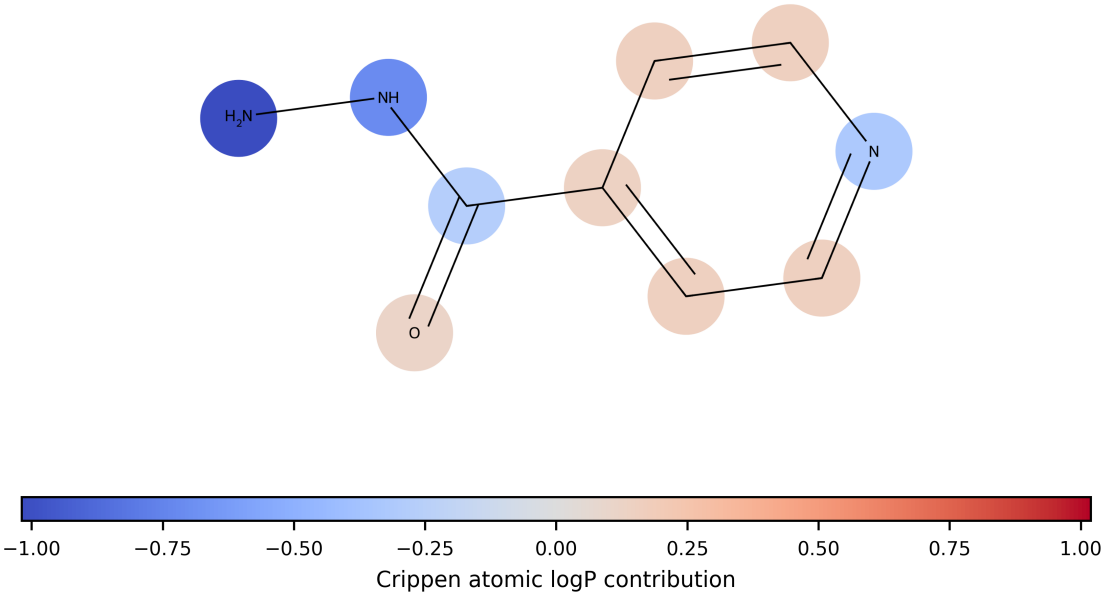


Figure S8: Fragment-level lipophilicity of two BBBP compounds: (A) zonisamide (BBBP #349, permeable) and (B) isoniazid (BBBP #279, non-permeable). Atoms are shaded by their Crippen logP contribution (blue = hydrophilic, red = lipophilic); the scale is shared by the two panels and is computed from the SMILES alone, so the figure is reproducible without the model. Both descriptors used in the text (Crippen logP and TPSA [7]) come from the SMILES alone; the similarity statistic of the scaffold splits uses ECFP4 [8].

5.3 Scale Reliance and Structural Validation

Table S11: Mean gate value per fusion stage, by endpoint type. A value near 1 means the first input of that stage dominates; a value near 0 means the second does.

Endpoint type	**Stage 1: atomic vs junction tree**	**Stage 2: fused vs functional group**
Biophysics / physiology	0.451	0.527
Physical chemistry	0.468	0.521
Quantum chemical	0.480	0.516

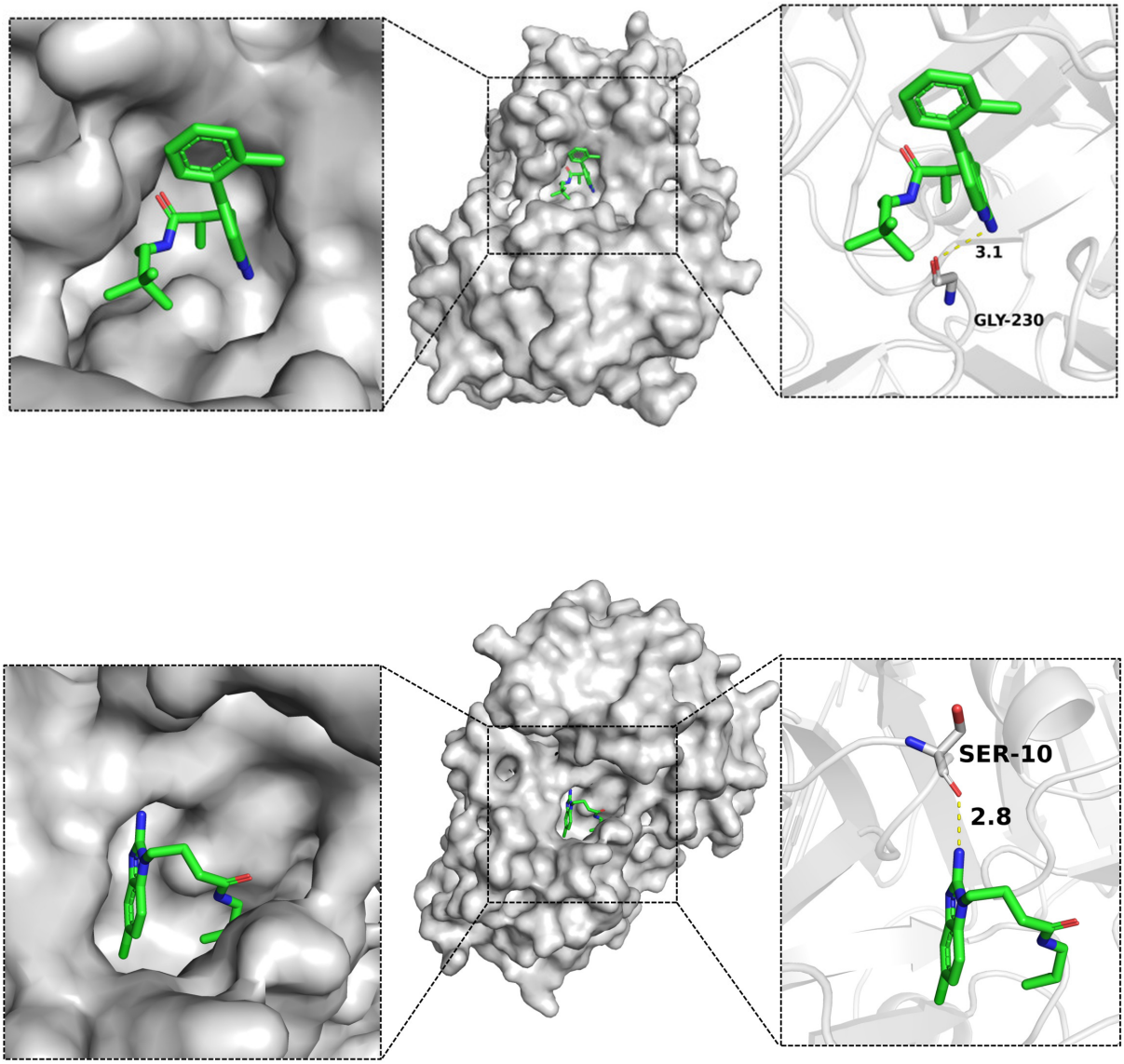


Figure S9: Docking validation on an active and an inactive BACE-1 inhibitor. Top row, BACE 276 (pIC50 = 9.18, active); bottom row, BACE 1547 (pIC50 = 2.54, inactive). The three panels show the binding-pocket surface, a solid surface map of the site, and the predicted 3D binding pose (AutoDock Vina [9], PDB 1FKN; rendered in PyMOL [10]).

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